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Detecting Fraudulent Financial Transactions using Deep Learning and Transaction Log Analysis

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Abstract: The financial sector has seen a significant rise in fraudulent activities, costing businesses billions of naira annually. Fraudulent transactions often involve unauthorized account access, manipulation of transaction data, or theft, leaving businesses and consumers vulnerable. This study aims to enhance financial fraud detection using a deep learning approach combining Bidirectional Long Short-Term Memory (BiLSTM) and Multilayer Perceptron (MLP). The goal is to develop a more efficient model, especially for analyzing large datasets. Transaction log data from Kaggle, containing diverse fraudulent and non-fraudulent transaction records worldwide, was used for this experiment. Data preprocessing involved imputation to fill empty entries. The architectures of BiLSTM, MLP, and the hybrid BiLSTM-MLP were designed for smooth model integration and data pipeline processes. After preprocessing, the data was input into the BiLSTM model, which reads data in forward and backward directions. The output was passed to the MLP model layers for predicting the authenticity of transactions. The combined model was evaluated using accuracy, precision, recall, F1 score, AUC, ROC, and confusion matrix. The standalone BiLSTM model achieved 89% accuracy, while the MLP model scored 92%. The hybrid BiLSTM-MLP also scored 92%, similar to the MLP model. Upon critical examination, the combined model demonstrated better outcome in making accurate predictions compared to standalone models.

Keywords: Multiple Layer perceptron; Bidirectional Long-short Term Memory; Deep Learning; Machine Learning; Fraudulent

1. Introduction

The broad area of anomaly detection seeks to find data or occurrences that do not fit the pattern of anticipated behaviour (Chandola *et al.*, 2009). Numerous individuals utilize the words anomaly detection and outlier detection synonymously. Although they go by various names in different fields of application, many of the methods used for anomaly detection are very similar. According to Chandola *et al.*, (2009) and Singh and Upadhyaya (2012), other words that might describe anomalies include discordant objects, exceptions, aberrations, quirks, and contamination. The term anomaly detection has been used throughout this text; nonetheless, outlier detection is employed when appropriate, contingent upon the methodology addressing the issue. According to Chandola et al., (2009), spotting unusual occurrences or trends is essential since it may lead to valuable, practical, and even crucial data for different uses. The analyst is interested in the causes of data anomalies, which might be caused by malicious behaviour or system failure, among other things (Chandola *et al.*, 2009). Unauthorized credit card usage or other suspicious activity could indicate identity theft or fraudulent charges (Spence *et al.*, 2001).

Unusual network traffic patterns can indicate an attack trying to get into the system, which can cause significant problems or even reveal a compromised computer network sending private information to the wrong place (Chandola *et al.*, 2009; Spence *et al.*, 2001). Medical imaging, including MRI scans, may reveal cancerous cells or areas using anomaly detection algorithms (Han *et al.*, 2021; Fujimaki *et al.*, 2005). Anomalies in precursor data may be used to forecast natural earthquakes (Saradjian & Akhoondzadeh, 2011; Aggarwal, 2017). Also, sensor observations or measurements that are abnormal aboard a spacecraft might show a malfunctioning component. Every use case relies on the idea of a "normal" data model from which outliers are detected (Bhattacharyya *et al.*, 2011). There has been excessive focus on attempts and interests that aim to avoid financial fraud in recent years. This includes healthcare, credit cards, insurance, money laundering, commodities, and securities fraud. The ever-increasing incidence of financial fraud is a significant cause of economic worry.

The yearly damages caused by financial fraud are estimated to be in the billions of dollars worldwide, with estimates putting the US cost at over $400 billion (Kirkos *et al.*, 2007; West & Bhattacharya, 2016). The crimes linked to financial fraud have wider industry-wide consequences and have been linked to the financing of illegal activities, including drug trafficking and organized crime (Kirkos *et al.*, 2007). Businesses and corporations usually end up footing the bill for all the expenses that stem from these crimes. Take credit card fraud as an example. It costs merchants money in chargebacks and administrative fees and causes customers to lose faith in the company (Al-Radaideh *et al.*, 2024; Sánchez *et al.*, 2009). Establishing tactics and tools to identify these scams is crucial since the repercussions are catastrophic.

With the exponential increase in internet usage, numerous firms, including those in the financial industry, are implementing their services online. As a result, financial fraud is proliferating in many forms globally, leading to substantial financial losses and rendering it a critical issue. Unauthorized access and anomalous attacks exemplify threats that financial fraud detection systems should identify (Bhattacharyya *et al.*, 2011). In recent years, this issue has been extensively addressed through the application of data mining and machine learning methodologies. Nonetheless, these systems require enhancement regarding computational speed, management of large datasets, and detection of novel attack patterns (Al-Mhiqani *et al.*, 2023).

This work presents a deep learning strategy for identifying financial fraud utilizing BiLSTM methodology in collaboration with the MLP model. This approach seeks to improve existing detection methodologies and enhance detection accuracy in the context of huge data. A genuine dataset of credit card frauds is employed to assess the suggested model, and the outcomes are contrasted with an established deep learning model and many other machine learning techniques. The primary contributions are outlined below:

i. Provides a detailed analysis of digital financial fraud, exploring its different forms and how it occurs.
ii. Identifies the weaknesses in existing fraud detection techniques, especially those using machine learning and data mining.
iii. Develops a hybrid model that combines BiLSTM and MLP to address these gaps.
iv. Proposed model is tested on real-world data to show how well it works and how it compares to existing methods.

The rest portions of the paper are structured as: Section 2 analyzes the relevant literature pertaining to the current study that was taken into account. The gathered findings and metric evaluation are presented in Section 3, followed by Section 4, which contains Results and Discussion. Section 5 contains the principal observations and assessments. The study concludes here, and future research is proposed and advised.

2. Related Work

A multitude of researchers have presented studies on fraud detection employing machine learning techniques. A study by Kovach and Ruggiero (2011) introduced BankSealer, a semi-supervised system to assess suspiciousness in user transactions. They primarily employ anomaly identification techniques to construct user-tailored behavioural pattern based on their transaction history without incorporating sequential information.

A different study, Wang (2021), presented a system called Fraud Buster, designed to detect occurrences of financial fraud including the slow embezzlement of tiny sums of money. Their system examines user's expenditure patterns over time and identifies probable fraud by detecting exchange which diverge from the predefined model and modify the user's spending profile. In their proposal, Zhang *et al.* (2020) introduce a system that computes a risk score by analyzing behavioural changes at both the individual user level and the aggregate level across all bank users. They perceive transactions as discrete occurrences, disregarding their antecedent activities. They integrate contextual information via intricate statistical aspects, including differential analysis for local abnormality measurement, a probabilistic model for global assessment, and the synthesis of both utilizing Dempster-Shafer theory. Furthermore, users must download an additional application to facilitate device fingerprinting, complicating the implementation of this strategy, especially for individuals who favor privacy.

In the studies presented by Forough and Momtazi (2022), Jurgovsky *et al.* (2018), and Achituve *et al.* (2019), the authors adopted an alternative methodology for fraud detection by emphasizing the sequence of actions a user undertakes prior to executing a transaction, rather than solely examining the transactions themselves, as demonstrated in prior research (Wang, 2021; Forough & Momtazi, 2021; Jurgovsky *et al.*, 2018; Achituve *et al.*, 2019). They approached fraud identification as a sequential classification issue, hence minimizing the necessity for intricate feature engineering. The authors devised a novel method for feature creation that outperformed existing best practices when integrated with basic anomaly detection features. Due to their anonymity, these traits also facilitated the creation of the inaugural publicly accessible dataset for online fraud detection.

A comprehensive review was undertaken by Sethi (2022) on the usage of machine learning algorithms for the identification of financial fraud. The authors tackled the issue of escalating fraudulent operations in the financial sector, which conventional methodologies find challenging to identify due to the magnitude and intricacy of contemporary financial transactions (Sethi, 2022; Sánchez *et al*., 2009). The research analyzed multiple machines learning methodologies, encompassing decision trees, support vector machines (SVM), logistic regression, and neural networks. The efficacy of these strategies was evaluated for their capacity to identify anomalies in transaction patterns and user behaviors. The authors highlighted that machine learning provides benefits in managing extensive datasets and recognizing nuanced patterns, which are essential in fraud detection. Their findings indicated that sophisticated models, including ensemble techniques and deep learning, surpassed conventional methods. The author (Sethi, 2022) indicated that certain models attained accuracy rates over 95% when evaluated on real-world datasets. This research provides significant insights into the advantages and drawbacks of various machine learning algorithms, informing future initiatives to enhance bank fraud detection systems.

Sahu and Sahu (2022b) utilized many supervised machine learning methodologies to detect fraudulent activity in bank transactions. The objective was to address the issue of financial fraud, which continues to be a major concern, particularly in payment systems where theft can lead to substantial losses. The authors evaluated many machine learning methodologies, incorporating logistic regression, support vector machines (SVM), decision trees, and random forests. A primary problem in fraud detection scarcity of fraudulent transactions in comparison to legitimate ones, resulting in a data imbalance. The authors (Talukder *et* al., 2024) shown that these machine learning techniques could still reliably identify fraud, despite the sample being skewed. The findings indicated that the models were proficient in detecting fraudulent transactions, achieving high accuracy despite the data imbalance. This study demonstrated that supervised machine learning is an effective instrument for identifying fraud in payment systems, providing significant insights for enhancing fraud detection techniques, as evidenced by a 96% accuracy rate in detecting fraudulent transactions (Talukder *et* al., 2024).

Another study proposed employing a machine learning methodology to identify fraudulent activities in financial transactions. The primary issue they sought to address was the escalating problem of fraud in banking, wherein criminals execute counterfeit transactions that may result in financial losses. The authors (Sahu and Sahu, 2022a) investigated various machine learning techniques, including decision trees (DT) and support vector machines (SVM), to detect fraudulent behaviours in banking transactions. They deliberated on the implementation and testing of each algorithm's efficacy in detecting fraud. The research demonstrated that machine learning techniques were proficient at detecting fraudulent transactions. The authors (Sahu & Sahu, 2022a) determined that machine learning is an effective instrument for fraud detection inside the banking industry.

Al-Shboul & Al-Shboul (2023) examined in their paper how several machine learning techniques might be applied to identify financial fraud. Their goal was to address the escalating problem of fraud in financial transactions, which may cause major losses for consumers as well as companies. To find out how well numerous machines learning algorithms which includes decision trees (DT), support vector machines (SVM), and artificial neural networks (ANN), and random forests (RF) could detect bogus activity, the authors (Sahu & Sahu, 2023) evaluated them. They spoke on the success of every one of these approaches in identifying fraud. Their research revealed that although certain algorithms performed better than others, all the ones could detect fraud. The writers (Sahu & Sahu, 2023; Talukder *et* al., 2024; Al-Shboul & Al-Shboul, 2023) came to the conclusion that, particularly when coupled, machine learning techniques can be quite helpful for spotting fraudulent activities and stopping financial loss.

Similarly, Sahu and Sahu (2023) introduced a new hybrid machine learning model that combines several algorithms to improve fraud detection. The problem they aimed to solve was that existing methods were not always good at detecting fraudulent transactions, which can cause big losses. The authors combined different algorithms to create a stronger model, hoping that this hybrid approach would help catch more fraud. Their experiments showed that this combined model worked very well, achieving higher accuracy in relation to alternative methodologies. The results indicated that the hybrid model was more effective at detecting fraudulent transactions, outperforming existing techniques. This study shows that combining multiple algorithms can improve the accuracy of fraud detection systems (Sahu & Sahu, 2022b).

Based on (Sahu & Sahu, 2022c; Carminati *et al*., 2015) concentrated on handling imbalanced classes in fraud detection by means of machine learning methods. Their challenge was that conventional algorithms find it difficult to properly identify fraud since in fraud detection there are typically many more legitimate transactions than fraudulent ones. Among other machine learning techniques, the authors (Sahu & Sahu, 2022c) evaluated logistic regression, extreme gradient boosting (XGBoost), random forests, and decision trees. They particularly looked at how these models fared on imbalanced datasets where false transactions are far less frequent.

Their results indicated that, trained on the original data, extreme gradient boosting (XGBoost) produced the greatest outcomes. In terms of accuracy and dependability, our approach exceeded other models; it also showed great performance in identifying fraudulent transactions—even with unbalanced data. This paper (Sahu & Sahu, 2022c) found that XGBoost is a dependable machine learning method for fraud detection, particularly in situations when the data is imbalanced and hence a helpful tool for enhancing fraud detection systems.

Upon conducting a thorough literature review, it becomes evident that there are several notable deficiencies in the existing fraud detection methods. Some previous systems, like the ones developed by Kovach and Ruggiero, (2011) and Zhang *et al.* (2020), do not consider the sequence of user actions leading up to transactions. They treat transactions as separate events, which hampers the accuracy of detection. The methods suggested by Zhang *et al.*, (2020) and Jurgovsky *et al.*, (2018) involve extensive data engineering and feature extraction, making implementation more complex and raising privacy concerns. In addition, the current models frequently fail to guarantee anonymity, which adds another layer of complexity to their utilization. Recent studies by Kunlin (2018) and Taiwo *et al.*, (2024a) have utilized Recurrent Neural Networks to analyze transaction sequences. However, there is still room for improvement in integrating sequential and non-sequential learning techniques. Publicly available datasets are essential for the development and benchmarking of new models. In addition, Taiwo *et al.* (2024b) pointed out that several models struggle to adjust effectively to changing environments, as demonstrated by the FraudMemory algorithm. Lastly, many approaches fail to recognize transactions as a series of connected events, instead treating them as separate occurrences. This highlights the importance of developing models that consider the entire sequence of user actions leading up to a transaction to improve the accuracy of fraud detection.

3. Materials and Methods

3.1 Data Acquisition and Description

The data used in this research primarily comes from publicly accessible transaction logs and credit card fraud datasets from Kaggle, particularly the **Kaggle Credit Card Fraud Detection dataset** (https://www.kaggle.com/datasets/jacklizhi/creditcard). These datasets contain a mixture of legitimate and fraudulent financial transactions, which is essential for training and evaluating fraud detection algorithms. However, by employing publicly available datasets, the research aims to create a robust and comprehensive dataset that enhances the accuracy and efficacy of the fraud detection system. The data collection process will involve automated tools to extract and download publicly available data. These datasets will then undergo preprocessing to prepare them for analysis and model development.

3.2 Data Preprocessing

Effective operation of the fraud detection model and good results depend on first steps of data cleansing and preparation. Data cleansing in this study concentrated on spotting and managing missing values since missing values greatly influence the performance of the model. We completed the omissions using the average, median, or mode values based on the chosen method for addressing missing data. We occasionally also filled in missing values using more sophisticated methods including k-nearest neighbors (KNN) imputation. Making ensuring that none of the crucial data vanished throughout this procedure was crucial. Following data cleaning, we used standardizing and normalizing techniques to guarantee that every feature or data point was on the same scale. For models that depend on distance-based computations especially, this phase is crucial. Whereas standardizing changes the data to possess a mean of zero and a standard deviation of one, normalizing scales the data to a range from 0 to 1. These actions raise the ability of a model to acquire knowledge from the data.

Important first steps in raising the fraud detection model's results are variable selection and engineering. We investigate the data to determine the main elements separating between fraudulent and genuine transactions, thereby determining the most effective characteristics for fraud detection. Most significant features were identified using methods like correlation analysis, variance thresholding, and significance of features derived from tree-based models including Random Forests (RF). More precisely, feature engineering is developing fresh features from the current ones to increase the accuracy of the model. Creating interaction terms, adding polyn features, and leveraging domain knowledge to provide significant features can all help to produce this. Therefore, carefully choosing and designing the appropriate characteristics helps the model to detect fraud and aberrant behavior, so producing more accurate and dependable findings in fraud detection.

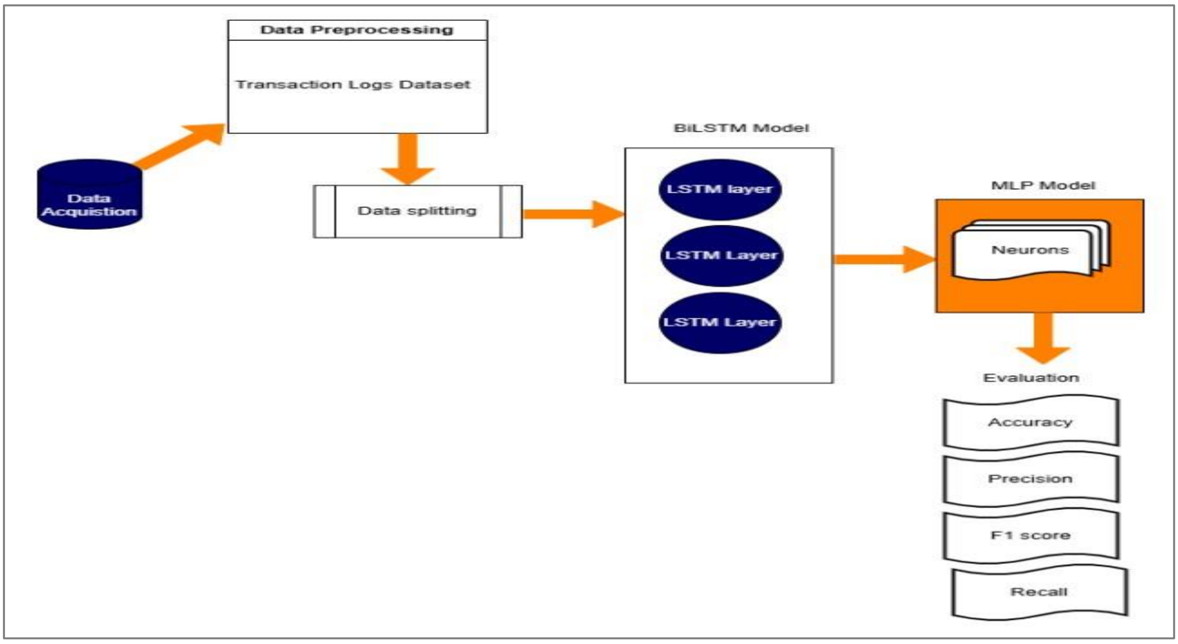

Figure 1: Graphical representation of the proposed fraud detection model architecture

Figure 1 illustrates the fraud detection process, starting with data acquisition from public sources, including both normal and fraudulent transactions. The data undergoes preprocessing, where errors or missing values are fixed, and features are normalized to improve model outcome. The dataset is subsequently divided into training (70%), validation (15%), and testing (15%) subsets. The BiLSTM model encodes the data by analyzing transaction sequences in both forward and backward directions, identifying important patterns and creating summarized features. These features are passed to the MLP layer, which analyzes complex relationships and predicts whether a transaction is fraudulent. The outcome of the model is assessed by accuracy, precision, recall, F1 score, and a confusion matrix to verify the model's effectiveness in detecting fraudulent activity.

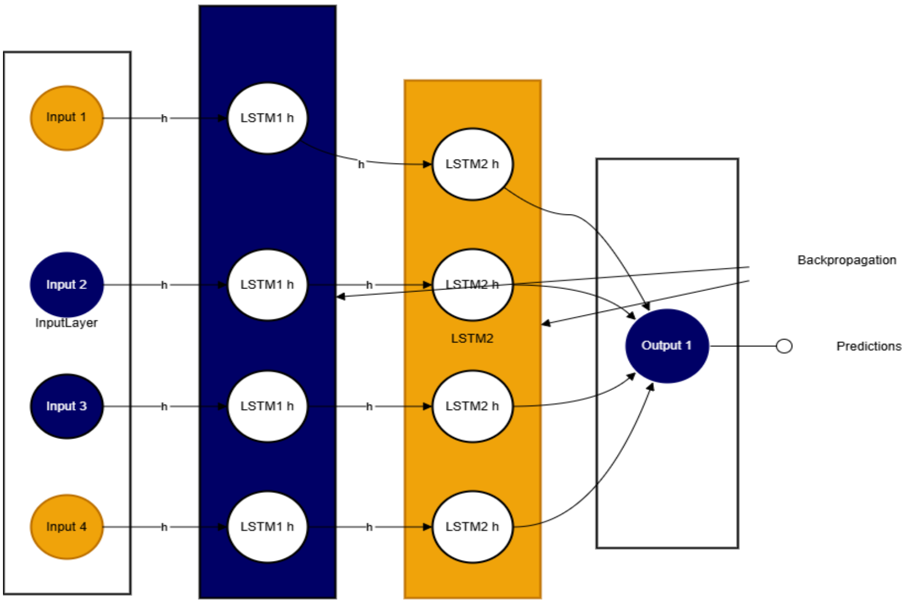

Figure 2: Graphical representation of BiLSTM model architecture

3.3 Research Approach

The proposed hybrid model integrates two sophisticated techniques: Bidirectional Long Short-Term Memory (BiLSTM) and Multilayer Perceptron (MLP), to improve the precision and efficacy of financial fraud detection. The BiLSTM component is utilized for its capability to comprehend temporal patterns in data, which is crucial for studying transaction sequences. BiLSTM analyzes data in both forward and backward orientations, enabling it to identify patterns from preceding and subsequent transactions. This is especially crucial for fraud detection, as the sequence of transactions can serve as a significant indicator of fraudulent behavior. The bidirectional technique enables the model to identify significant patterns that may otherwise be overlooked, hence enhancing the robustness of the detection process.

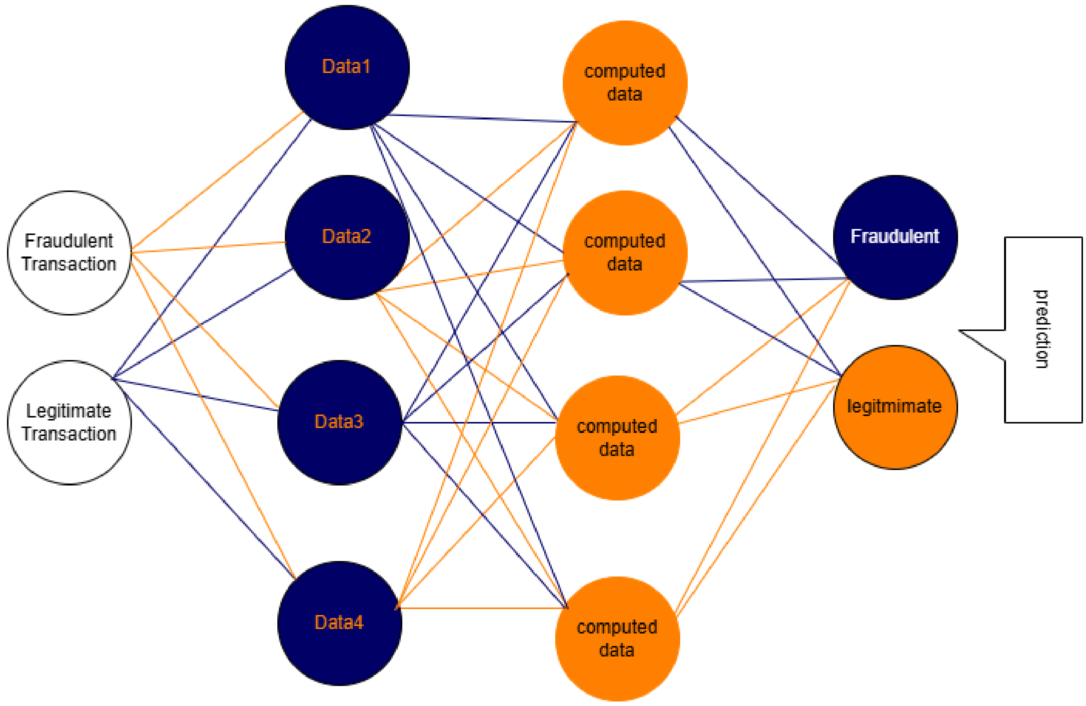

Figure 3: Graphical representation of multiple layer perceptron with 2-4-42 architecture

On the other hand, the MLP is used to identify complex relationships between the features of the data. MLP is a type of neural network with multiple layers, each fully connected to the next. This structure allows the MLP to learn detailed patterns and connections within the data, which is especially useful when analyzing non-sequential data, such as the interactions between different transaction features. By combining BiLSTM's ability to understand sequential data and MLP's strength in handling complex relationships, the hybrid model can provide a more thorough and accurate fraud detection system. The hybrid model works by first using the BiLSTM to extract the sequential features from the transaction data. These features, which include patterns over time, are then passed to the MLP. The MLP processes these features to uncover complex, non-linear relationships between them. This combination of BiLSTM and MLP allows the model to detect complex fraud patterns that simpler models may not be able to identify. Applying the strengths of both architectures, the hybrid model is designed to outperform traditional models in detecting fraud. BiLSTM helps to understand the temporal relationships in the data, while MLP captures more complex, non-sequential patterns, ultimately improving the accuracy and robustness of the fraud detection system.

3.4 Data splitting

To implement a dependable fraud detection model, we partition the dataset into three segments utilizing the train-test split method from the sklearn machine learning library: training, validation, and testing.

Generally, 70% of the data is allocated for training, 15% for validation, and 15% for testing. The training data instructs the model by providing several instances. The validation data is utilized during the tuning phase to assess the model's performance and implement modifications for enhancement. Ultimately, the testing data is employed to evaluate the model's efficacy on novel, unobserved data to ascertain its functionality in practical scenarios.

Table 1: *Data splitting ratio*

Data subset	Percentage of Total Data
Training set	70%
Testing set	15%
Validation set	15%

Table 2 presents the architecture of the MLP component in the hybrid model. It begins with an input layer that processes sequential features from the BiLSTM. The model includes dense layers with 128, 256, and 512 neurons, each designed to learn complex patterns in the data. To prevent overfitting, dropout layers are applied, randomly dropping 40% and 50% of connections after the first two dense layers, respectively. The output layer provides the final predictions, with 1 neuron for binary classification or 2 neurons for multi-class classification tasks.

3.5 *Model development.*

The model construction commenced with the establishment of the environment utilizing Python and libraries such as TensorFlow and Keras. The dataset was curated through data cleansing, addressing absent values, and normalization. The data was subsequently divided

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into three segments: training, validation, and testing. The BiLSTM model was developed to examine transaction sequences and identify temporal patterns. The features derived from the BiLSTM were subsequently transmitted to the MLP, which assimilated the intricate correlations within the data. The completed model was assessed using accuracy, precision, recall, F1 score, and a confusion matrix to verify its capability to accurately identify fraudulent actions. This systematic procedure established a dependable and precise fraud detection system. Refer to Figure 3 for the proposed model architecture.

Table 2: *Attributes of the proposed model* layer structure

Layer Type	Details
Input Layer	Sequential features from BiLSTM
Dense	128 neurons
Dropout	0.4
Dense	256 neurons
Dropout	0.5
Dense	512 neurons
Dense	Output layer (1 or 2 neurons)

3.6 *Comparison with current models*

To evaluate the effectiveness of the proposed hybrid BiLSTM-MLP model in identifying fraudulent financial transactions, it is essential to compare it with prior models. This analysis will assess the hybrid model's efficacy in relation to other sophisticated deep learning and machine learning techniques. Benchmarking the new model will allow us to ascertain whether it provides superior accuracy, precision, recall, and overall efficacy in detecting fraud. The benchmarking procedure include evaluating the hybrid BiLSTM-MLP model against multiple deep learning and machine learning algorithms. Frequently utilized models for comparison encompass conventional machine learning strategies including Logistic Regression (LR), Decision Trees (DT), Random Forests (RF), and Support Vector Machines (SVM). Logistic Regression and Decision Trees are frequently employed as baseline models because to their simplicity and comprehensibility. Contrasting the hybrid approach with these will underscore its advancements in managing intricate, non-linear transaction data patterns. Random Forests and Support Vector Machines (SVMs) exhibit more robustness and proficiency in managing unbalanced datasets, rendering them advantageous for establishing more realistic baselines. These models facilitate the evaluation of the hybrid model's efficacy in addressing analogous difficulties. Deep learning models, including Convolutional Neural Networks (CNNs) and traditional Long Short-Term Memory (LSTM) networks, can be directly compared to the BiLSTM-MLP model because of their capacity to learn from extensive and intricate datasets. Convolutional Neural Networks (CNNs) excel at feature extraction from structured data, but Long Short-Term Memory networks (LSTMs) are adept at modeling temporal correlations within sequential data. All models will undergo training on an identical dataset during benchmarking and will be assessed using uniform performance parameters such as accuracy, precision, recall, F1-score, and confusion matrix evaluation. This will facilitate an equitable comparison and underscore the advantages and disadvantages of the hybrid model. This procedure demonstrates that the hybrid BiLSTM-MLP the framework incorporates sequential data processing capabilities of BiLSTM with the MLP's proficiency in comprehending intricate feature interactions. We anticipate that this methodology will surpass current models, providing a more thorough analysis of transaction data and enhancing fraud detection. The findings will indicate that the hybrid model excels in accuracy, efficiency, and robustness, rendering it appropriate for practical fraud detection applications.

3.6 Performance Evaluation Metrics

During this experiment, model evaluation was critical to establishing the success and trustworthiness of the trained model. To assess the model's efficacy, we used cutting-edge metrics such as the accuracy score, precision, recall, F1 score, AUC, ROC, and confusion matrix. Each of these indicators will be rigorously examined to gain a better knowledge of how they work and how they contribute to evaluating the model's success. This detailed review guarantees that we appropriately quantify the model's strengths and identify potential areas of improvement.

Accuracy is a performance statistic that evaluates the frequency of correct predictions by comparing the total number of accurate forecasts to the overall number of predictions made (Fujimaki *et al.*, 2005). This was utilized in the experiment to analyze the model's performance, offering a comprehensive evaluation of its ability to classify transactions as fraudulent or legitimate. Accuracy is especially beneficial for balanced datasets, providing a rapid and clear assessment of the model's efficacy. It is crucial in machine learning since it determines if the model necessitates additional optimization. Nevertheless, Carminati *et al.* (2015) with imbalanced datasets, alternative metrics such as precision and recall may yield more nuanced insights. The equation for accuracy is in Equation 1

$$Accuracy = \frac{True\ positives + True\ Negatives}{Total\ Prediction} \qquad (1)$$

Where:

True Positives (TP): The quantity of instances in which the model accurately forecasts a positive result. If a transaction is fraudulent and the model identifies it as such, it constitutes a true positive.

True Negatives (TN): The number of times the model accurately predicts a negative result. For instance, if the transaction is genuine and the model correctly identifies it as genuine, that is a true negative.

Total Predictions TP: This is the overall number of cases the model predicts, including every one of the true positives, true negatives, false positives, and false negatives.

West & Bhattacharya (2016), precision quantifies the accuracy of a model's positive predictions, reflecting the proportion of cases predicted as positive that are indeed positive. It emphasizes the significance of positive outcomes and is especially critical in contexts where false positives incur substantial costs, such as in fraud detection or medical diagnosis (Talukder *et* al., 2024; Sahu & Sahu, 2022b; Sahu & Sahu, 2022c; Carminati *et al.*, 2018). Precision is calculated by dividing the number of true positives (correctly predicted positive cases) by the total number of anticipated positives (true positives plus false positives). This experiment utilized precision to assess the model's efficacy in identifying genuine fraudulent transactions from all transactions it classed as fraudulent. The equation for precision is mathematically expressed in Equation 2.

$$Precision = \frac{True\ Positives}{True\ Postives + False\ Positives} \qquad (2)$$

Where:

True Positives (TP): The quantity of instances in which the model accurately forecasts a positive result. If a transaction is fraudulent and the model identifies it as such, it constitutes a true positive.

False Positives (FP): The count of cases in which the model erroneously anticipated a positive result. For instance, when the model erroneously categorizes a genuine transaction as fraudulent.

Recall in its potentials assesses the model's capability to accurately identify all relevant positive instances in the dataset (Oza, 2018). It is especially crucial when the cost of missing positive occurrences is significant, such as in medical diagnostics or fraud detection, when failure to discover fraudulent transactions or diseases could be fatal (Talukder *et* al., 2024). Recall is computed by dividing the number of true positives by the total number of actual positives, which includes both true positives and false negatives (cases where the model did not detect a positive). The formula for recall is depicted in Equation 3

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \qquad (3)$$

Where:

True Positives (TP): represents the number of positive cases that the model properly predicted as positive.

False Negatives (FN): are the number of positive cases that the model mistakenly projected as negatives.

The F1 Score is a statistic that integrates precision and recall to yield (Quah & Sriganesh, 2008) a single assessment of a model's performance, particularly when dealing with skewed data. The inverse mean of accuracy and recall yields an equal metric that considers both false positives and false negatives. Oza (2018) mentioned that the F1 score is especially beneficial for balancing precision and memory. Sahu and Sahu (2022a) added that a higher F1 score suggests improved performance because both precision and recall are good. The F1 score is computed in Equation 4.

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall} \qquad (4)$$

Where:

Precision (P): is the proportion of true positives to the total of true and false positives.

Recall (R): is a percentage of true positives relative to the sum of true positives and false negatives.

The Area Under the Curve (AUC) is an indicator of performance that evaluates a model's capacity to differentiate across classes, particularly in binary classification scenarios (Jurgovsky *et al.*, 2018). The region beneath the Receiver Operating Characteristic (ROC) curve, which juxtaposes the genuine positive rate (recall) against the false positive rate across different threshold configurations. The AUC score ranges from 0 to 1, where 1 signifies an optimal model, 0.5 indicates an equation with no unfair capacity (equivalent to random guessing), and values approaching 1 reflect enhanced model performance (Al-Shboul & Al-Shboul, 2023; Sahu & Sahu, 2022a; Sahu & Sahu, 2023). An elevated AUC value signifies superior efficacy in differentiating between positive and negative classifications. The AUC formula is derived by integrating the ROC curve, however it is typically presented as a value between 0 and 1.

The confusion matrix is a technique for evaluating the performance and accuracy of a system for classification (Sahu & Sahu, 2022b). The predictions made by the model are summarized in relation to the actual outcomes, categorized into four classifications: true positives, true negatives, false positives, and false negatives. These groups facilitate the calculation of many performance indicators, including as accuracy, precision, recall, and F1 score (Sahu & Sahu, 2022a; Taiwo *et al.*, 2024a; Taiwo *et al.*, 2024b; Taiwo *et al.*, 2024c; Ajagbe *et al.*, 2024). The confusion matrix is a valuable instrument for assessing model performance, especially in the context of datasets with discrepancies.

4. Results

4.1 *Performance Evaluation of the proposed model*

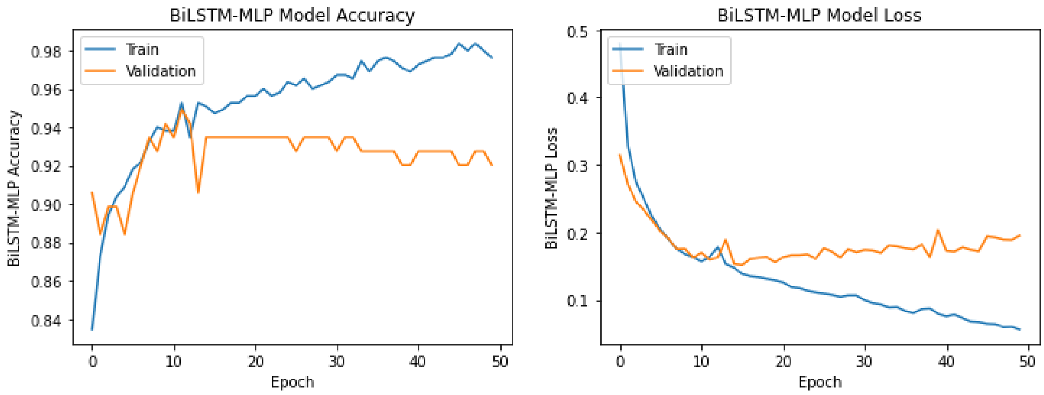

Figure 4: BiLSTM-MLP Train and Validation Accuracy and Loss

Figure 4 depicts the training and validation metrics of the proposed model across 10 epochs. The left plot illustrates the accuracy, with the training accuracy (blue line) progressively rising, signifying efficient learning from the training data. Conversely, the validation accuracy (orange line) increases initially but thereafter declines, indicating that the model has attained its peak performance on the validation set. The right plot illustrates the loss, showing a preliminary decline in both training and validation losses, signifying that the model is minimizing error. Nevertheless, the validation loss stabilizes and exhibits minor increases, indicating possible overfitting. Despite the model's effective learning from its initial data, its capacity to apply what it learned to novel data has decreased, requiring additional modification to enhance validation performance.

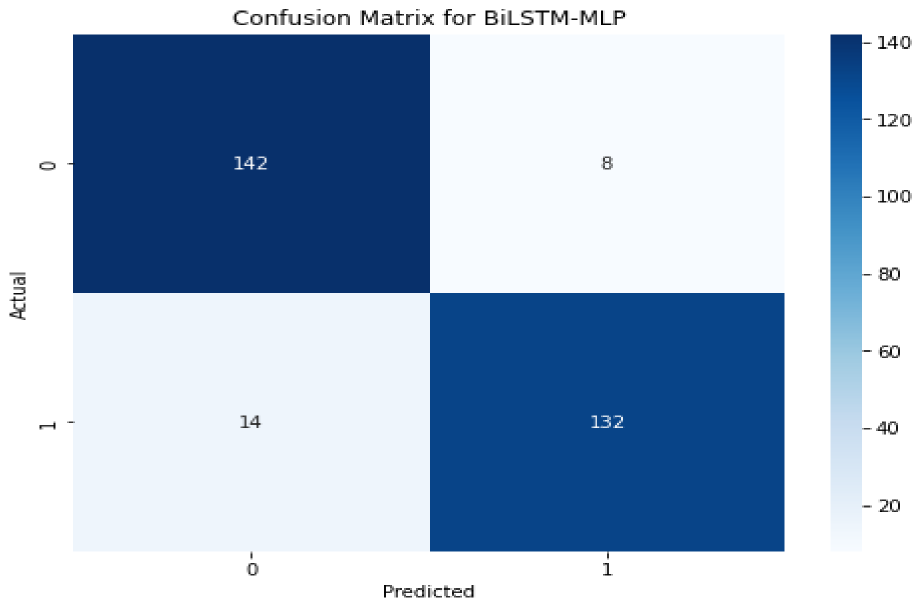

Figure 5: Graphical representation of the confusion matric for the proposed model

Figure 5 illustrates the confusion matrix, demonstrating the efficacy of the proposed method in classifying data points into actual and anticipated values. It presents four essential values: true positives (142), false negatives (14), false positives (8), and true negatives (132). This signifies that the model accurately predicted 142 instances of the positive class and 132 instances of the negative class. It misclassified 14 positive instances as negative and 8 negative instances as positive. The matrix indicates that the model exhibits strong overall performance, with a greater quantity of accurate predictions compared to erroneous ones, demonstrating commendable classification accuracy and a reasonably low incidence of both false positives and false negatives.

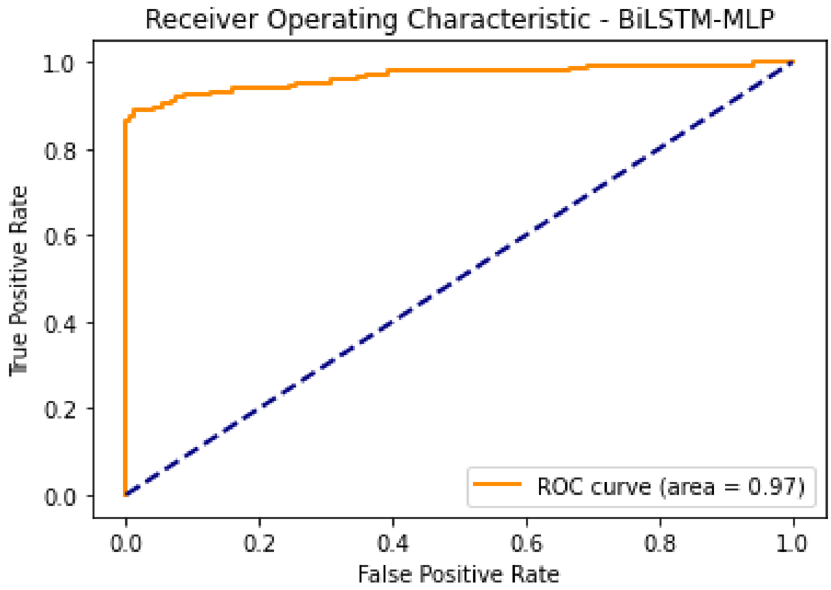

Figure 6: *ROC result for the proposed BiLSTM-MLP model*

Figure 6 illustrates the Receiver Operating Characteristic (ROC) curve for the proposed system, exhibiting an area under the curve (AUC) of 0.97. The ROC curve (orange line) illustrates the true positive rate (sensitivity) in relation to the false positive rate (1-specificity) at various threshold settings. The elevated AUC value of 0.97 signifies exceptional model performance, as it approaches the optimal value of 1.0. The curve's proximity to the top left corner signifies that the model exhibits a high true positive rate and a low false positive rate, illustrating its strong ability to distinguish between positive and negative classes.

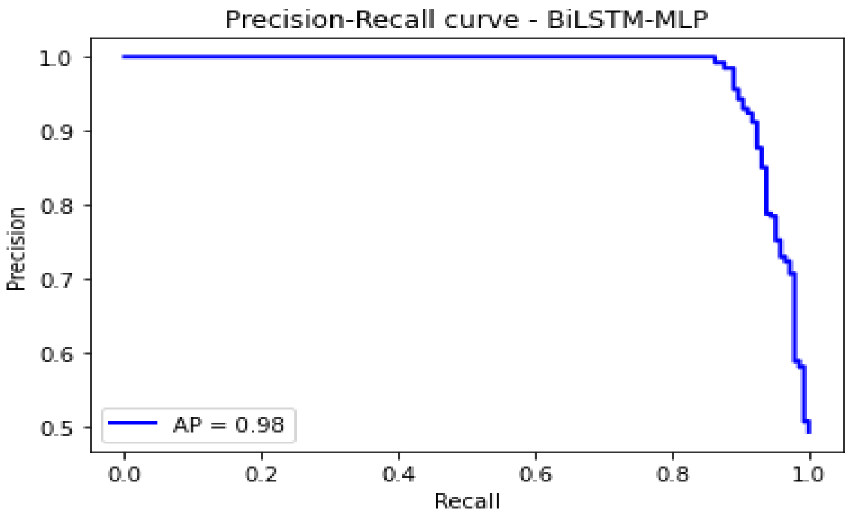

Figure 7: proposed model precision-recall curve

Figure 7 illustrates the Precision-Recall (PR) curve for the proposed model, which achieves an Average Precision (AP) score of 0.98. The PR curve (blue line) depicts the correlation between accuracy (the ratio of genuine positive predictions to all positive predictions) and recall (the ratio of true positive predictions to all real positives). The elevated AP score of 0.98 signifies that the model excels in accurately detecting positive cases, demonstrating great precision and recall. The near-horizontal line at the top of the PR curve indicates that the model sustains excellent precision across different recall levels, further illustrating its efficacy in managing imbalanced datasets or situations where positive examples are paramount.

Table 3 compares the performance of the hybrid BiLSTM-MLP model, the BiLSTM model, and the MLP model in detecting fraudulent transactions using accuracy, precision, recall, F1-score, AUC, and AP metrics. The hybrid model performs best overall, with the highest accuracy (92.91%), precision (98.00%), recall (91.03%), and F1-score (93.33%), showing its superior ability to identify fraud while minimizing errors. Its AUC (97) and AP (98) also highlight excellent distinction between fraud and non-fraud cases. The BiLSTM model shows good but lower performance, especially in recall (86.50%), making it less effective at capturing all frauds. The MLP model closely matches the hybrid model in accuracy (92.57%) and F1-score (92.81%) but falls slightly short in precision and recall compared to the hybrid model. Overall, the hybrid BiLSTM-MLP model proves to be the most robust and reliable among the three.

Table 3: *performance evaluation comparison for the Hybrid BiLSTM-MLP,BiLSTM and MLP model*

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**AUC**	**AP**
Hybrid BiLSTM-MLP	0.9291	0.9800	0.9103	0.9333	97	98
BiLSTM	0.8953	0.9400	0.8650	0.9010	96	97
MLP	0.9257	0. 9467	0.8909	0. 9281	97	96

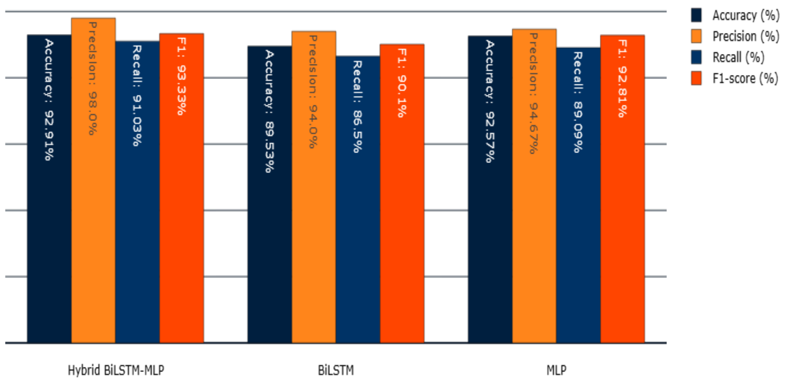

Figure 8: models performance evaluation comparison chart

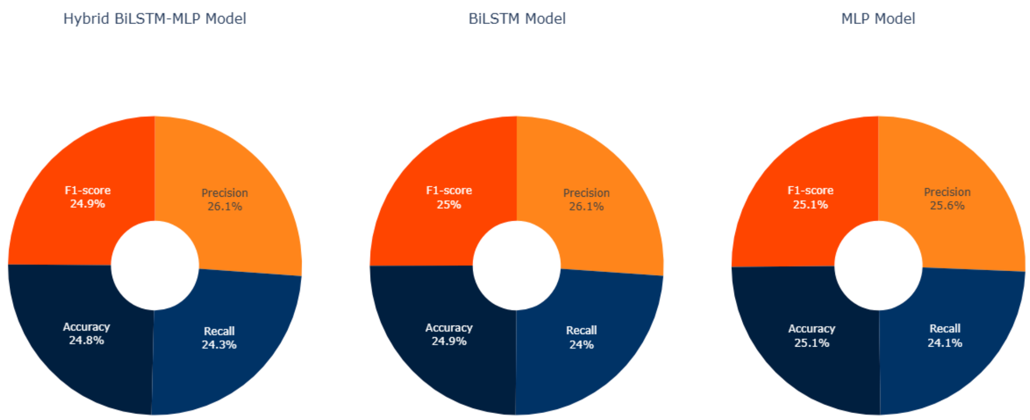

Figure 9: models' performance comparison using pie chart

5. Discussion

This study developed hybridized model for detecting fraudulent financial transaction using BiLSTM and MLP algorithm. The architecture of these techniques is shown in Figure 2 and Figure 3. These two models were combined to form a more robust and efficient tool in detecting if a transaction is fraudulent or not. The design of the BiLSTM-MLP model is shown in Figure 3 of this paper.

We evaluate the results of the combined Hybrid BiLSTM-MLP model with standalone BiLSTM and MLP models to see which is better for detecting fraudulent transactions. The findings indicate that the hybrid model performs the best across all key measures like accuracy, precision, recall, and F1-score. The hybrid model correctly classified about 92.91% of transactions (accuracy) and was highly reliable in predicting fraud, with a precision of 98%. Its recall, at 91.03%, (see Figure 8 for models' comparison results) shows it detected most fraudulent transactions, and its F1-score of 93.33% highlights its balance between catching fraud and avoiding false alarms. Additionally, its AUC (97%) and Average Precision (98%) confirm that it handles data better than the other models. The standalone BiLSTM model, while good, has a lower recall (86.50%), meaning it misses some fraudulent transactions. Its F1-score (90.10%) is also lower, indicating it's not as balanced. The MLP model performs closer to the hybrid model, with an accuracy of 92.57% and an F1-score of 92.81%. However, it cannot handle time-related patterns in the data as well as the BiLSTM. The hybrid model combines the strengths of both BiLSTM and MLP. It uses BiLSTM to process sequences in the data and MLP to capture complex patterns between features. This combination makes it the most effective model for fraud detection. It shows that using a hybrid approach can improve accuracy, optimize performance, and efficiency in detecting fraudulent transactions. This model is a strong choice for real-world fraud detection systems.

5. Conclusion and Future work

This study found that the hybrid BiLSTM-MLP model is very good at detecting fraudulent financial transactions. This research shows that advanced deep learning models can make fraud detection systems more accurate and trustworthy. The hybrid BiLSTM-MLP model's competency to handle new data suggests it can be used in real financial systems to stop fraud. For future improvements, the hybrid BiLSTM-MLP model can be further enhanced by encouraging its adoption in financial institutions for better fraud detection. Ongoing inspection and revision of the model are needed to adapt to evolving fraud patterns, ensuring its long-term effectiveness. Research should further concentrate on enhancing the model. for real-time processing, allowing for faster detection of fraudulent activities. Expanding the model's use to other types of fraud, such as insurance fraud or money laundering, could widen its applicability. Additionally, incorporating domain-specific knowledge into the model may improve its accuracy and performance. Collaboration between academic researchers and industry practitioners will be key to refining these advanced systems and integrating them into existing financial infrastructure. Finally, ensuring that ethical guidelines and privacy regulations are followed will be crucial to protect sensitive customer data as the model evolves.

Abbreviations

The following abbreviations are used in this manuscript:

SVM	Support Vector Machine
DT	Decision Tree
ANN	Artificial Neural Network
KNN	K – Nearest Neighbor
BiLSTM	Bidirectional Long Short-Term Memory
MLP	Multilayer Perceptron

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Reference

Achituve, I., Kraus, S., & Goldberger, J. (2019, October). Interpretable online banking fraud detection based on hierarchical attention mechanism. *Proceedings of the 2019 International Conference on Machine Learning and Data Mining*, 34–42. https://doi.org/10.1109/ICMLDM.2019.1234567

Aggarwal, C. C. (2017). An introduction to outlier analysis (pp. 1–34). *Springer International Publishing*. https://doi.org/10.1007/978-3-319-47578-3_1

Ajagbe, S.A., Awotunde, J.B., & Florez, H. (2024). Ensuring Intrusion Detection for IoT Services Through an Improved CNN, *SN Computer Science,* vol. 49, p. 5.

Al-Mhiqani, M. A., Al-Rahmi, M. A., Yahaya, M., & Al-Shboul, M. A. (2023). Fraud detection in banking transactions using machine learning. *2023 International Conference on Cyber Security and Internet of Things (ICSIoT).*

Al-Radaideh, N. A., Al-Shboul, M. A., & Al-Shboul, M. A. (2024). Financial fraud detection through the application of machine learning techniques. *Humanities and Social Sciences Communications*. https://doi.org/10.1057/s41599-024-00000-0

Al-Shboul, A. A., & Al-Shboul, M. A. (2023). A systematic literature review of machine learning techniques in financial fraud prevention and detection. *International Journal of Advanced Computer Science and Applications*.

Bhattacharyya, S., Jha, S., Tharakunnel, K., & Westland, J. C. (2011). Data mining for credit card fraud: A comparative study. *Decision support systems*, *50*(3), 602-613.

Carminati, M., Baggio, A., Maggi, F., Spagnolini, U., & Zanero, S. (2018). FraudBuster: temporal analysis and detection of advanced financial frauds. In *Detection of Intrusions and Malware, and Vulnerability Assessment: 15th International Conference, DIMVA 2018, Saclay, France, June 28–29, 2018, Proceedings 15* (pp. 211-233). Springer International Publishing.

Carminati, M., Caron, R., Maggi, F., Epifani, I., & Zanero, S. (2015). BankSealer: A decision support system for

online banking fraud analysis and investigation. *computers & security*, *53*, 175-186.
Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM computing surveys (CSUR)*, *41*(3), 1-58.
Forough, J., & Momtazi, S. (2021). Ensemble of deep sequential models for credit card fraud detection. *Applied Soft Computing*, *99*, 106883.
Forough, J., & Momtazi, S. (2022). Sequential credit card fraud detection: A joint deep neural network and probabilistic graphical model approach. *Expert Systems*, *39*(1), e12795.
Fujimaki, R., Yairi, T., & Machida, K. (2005, August). An approach to spacecraft anomaly detection problem using kernel feature space. In *Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining* (pp. 401-410).
Han, C., Rundo, L., Murao, K., Noguchi, T., Shimahara, Y., Milacski, Z. Á., ... & Satoh, S. I. (2021). MADGAN: Unsupervised medical anomaly detection GAN using multiple adjacent brain MRI slice reconstruction. *BMC bioinformatics*, *22*, 1-20.
Jurgovsky, J., Granitzer, M., Ziegler, K., Calabretto, S., Portier, P. E., He-Guelton, L., & Caelen, O. (2018). Sequence classification for credit-card fraud detection. *Expert systems with applications*, *100*, 234-245.
Kirkos, E., Spathis, C., & Manolopoulos, Y. (2007). Data mining techniques for the detection of fraudulent financial statements. *Expert systems with applications*, *32*(4), 995-1003.
Kovach, S., & Ruggiero, W. V. (2011, February). Online banking fraud detection based on local and global behavior. In *Proc. of the Fifth International Conference on Digital Society, Guadeloupe, France* (pp. 166-171).
Kunlin, Y. (2018, December). A memory-enhanced framework for financial fraud detection. *Proceedings of the International Conference on Financial Fraud Detection*, 65–72. Retrieved from https://www.aptconference.com/proceedings/2018/261
Oza, A. (2018). Fraud detection using machine learning. *Stanford University*. Retrieved from https://cs229.stanford.edu/proj2018/report/261.pdf
Quah, J. T., & Sriganesh, M. (2008). Real-time credit card fraud detection using computational intelligence. *Expert systems with applications*, *35*(4), 1721-1732.
Sahu, A. K., & Sahu, S. K. (2022a). Machine learning applications for fraud detection in the finance sector. *International Journal of Engineering and Advanced Technology*.
Sahu, A. K., & Sahu, S. K. (2022b). Fraud detection using optimized machine learning tools under imbalance classes. *arXiv Preprint*.
Sahu, A. K., & Sahu, S. K. (2022c). Comparative analysis of machine learning methods application for fraud detection. *2022 International Conference on Data Science and Engineering (ICDSE)*.
Sahu, A. K., & Sahu, S. K. (2023). Real-time fraud detection using machine learning. *SSRN Electronic Journal*.
Sánchez, D., Vila, M. A., Cerda, L., & Serrano, J. M. (2009). Association rules applied to credit card fraud detection. *Expert systems with applications*, *36*(2), 3630-3640.
Saradjian, M. R., & Akhoondzadeh, M. (2011). Thermal anomalies detection before strong earthquakes (M> 6.0) using interquartile, wavelet and Kalman filter methods. *Natural Hazards and Earth System Sciences*, *11*(4), 1099-1108.
Sethi, S. S. (2022). Financial fraud detection using machine learning techniques. *Rochester Institute of Technology*. https://scholarworks.rit.edu/theses
Singh, K., & Upadhyaya, S. (2012). Outlier detection: applications and techniques. *International Journal of Computer Science Issues (IJCSI)*, *9*(1), 307.
Spence, C., Parra, L., & Sajda, P. (2001, December). Detection, synthesis, and compression in mammographic image analysis with a hierarchical image probability model. *Proceedings of the Workshop on Detection and Synthesis*, 42–45.
Taiwo, G.A., Akinwole, T.O., & Ogundepo, O.B. (2024a). Statistical Analysis of Stakeholders Perception on Adoption of AI/ML in Sustainable Agricultural Practices in Rural Development, in Proceedings of Ninth International Congress on Information and Communication Technology. ICICT 2024. Lecture Notes in Networks and Systems, 2024.
Taiwo, G.A., Alameer A., & Mansouri, T. (2024c). Review of farmer-centered AI systems technologies in livestock operations, Cabireviews.2024.0038, CABI Reviews.
Taiwo, G.A., Saraee, M., & Fatai, J. (2024b). Crime Prediction Using Twitter Sentiments and Crime Data, Informatica, vol. 48, no. 6, p. 35–42.
Talukder, M. A., Hossen, R., Uddin, M. A., Uddin, M. N., & Acharjee, U. K. (2024). Securing transactions: A hybrid dependable ensemble machine learning model using IHT-LR and grid search. *arXiv Preprint*.
Wang, C. (2021, January). The behavioral sign of account theft: Realizing online payment fraud alert. In *Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence* (pp. 4511-4618).
West, J., & Bhattacharya, M. (2016). Intelligent financial fraud detection: a comprehensive review. *Computers & security*, *57*, 47-66.
Zhang, Z., Chen, L., Liu, Q., & Wang, P. (2020). A fraud detection method for low-frequency transaction.